

DBA-MA-UxS-NA

Uncrewed Systems: Nuclear Sector Implementation Annex

*An Implementation Annex to DBA-UxS: Cross-Sector
Foundational Document*

Applicable to All National Implementations of the DBA-MA Series

Parent Document: DBA-MA International Framework (L1, v1.4)

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This document applies to all national implementations of the DBA-MA framework.

Revision History

Version	Date	Author	Description
1.0	April 2026	T. Roxey	Initial release. Establishes the nuclear sector implementation annex to DBA-UxS v1.0. Migrates and consolidates all nuclear-specific content from DBA-MA-UxS v1.0: safety-significant system definitions, nuclear facility vulnerability profiles (UAS, USV, UUV), Autonomous Safe Shutdown Condition (ASSC) integration, nuclear-specific hardening requirements, parent document cross-references (L1–L4, TA), and IAEA references. DBA-MA-UxS v1.0 is superseded by the combination of DBA-UxS v1.0 and this document.
1.1	April 2026	Tim Roxey	Transitional note regarding DBA-MA-UAS TA reference updated to reflect completion of DBA-MA Transition Architecture v1.1, which replaced all DBA-MA-UAS references with DBA-UxS and DBA-MA-UxS-NA citations.

A1. Purpose and Scope of this Annex

This document is the nuclear sector implementation annex to DBA-UxS v1.0, the cross-sector foundational document for uncrewed systems threat considerations under the Design Basis Accident methodology. It is a child document of DBA-UxS and shall be read in conjunction with that document. Content established in DBA-UxS — including the UxS taxonomy, common threat characteristics, generic threat characterizations, sector-agnostic hardening design principles, and review cycle requirements — is not repeated here. This annex addresses only the nuclear-sector-specific elements that are required to complete the DBA-UxS framework for nuclear power plants and supporting nuclear infrastructure.

This annex applies to all national implementations of the DBA-MA series and supersedes the nuclear-specific content of DBA-MA-UxS v1.0 in conjunction with DBA-UxS v1.0. The designator DBA-MA-UxS v1.0 is retired; all references in the DBA-MA series documents to DBA-MA-UxS (any version) shall be understood as referring to the combination of DBA-UxS v1.0 and this annex upon adoption of both documents by the applicable national implementation.

This annex addresses, for the nuclear sector:

Section A2: Nuclear safety-significant systems, structures, and components — the definition of the target set for UxS threat characterization in the nuclear context.

Section A3: Autonomous Safe Shutdown Condition (ASSC) — the nuclear protected operational state, its definition, phases, and relationship to UxS hardening requirements.

Section A4: Nuclear facility vulnerability profiles for UAS, USV, and UUV — the application of the generic vulnerability profiles from DBA-UxS Sections 4.2, 5.2, and 6.2 to nuclear facility systems and designs.

Section A5: Nuclear-specific consequence chains — the consequence chain from UxS strike to radiological release threshold, including intermediate states.

Section A6: Nuclear-specific hardening requirements — the application of DBA-UxS Section 8 hardening principles to nuclear SSCs, with ASSC-referenced design requirements.

Section A7: Cross-domain compounding in the nuclear context — nuclear-specific development of the sector-specific compounding scenarios called for in DBA-UxS Section 7.4.

Section A8: Relationship to DBA-MA parent documents — cross-references to the L1, L2, L3, L4, and TA documents.

Section A9: Nuclear-specific references — IAEA standards, national regulatory instruments, and DBA-MA series documents.

A2. Nuclear Safety-Significant Systems, Structures, and Components

For purposes of this annex, nuclear safety-significant systems, structures, and components (SSCs) are those required to perform or support the three fundamental nuclear safety functions: (1) control of reactivity; (2) cooling of the reactor core and the spent fuel pool; and (3) confinement of radioactive material. These three functions define the boundary of the nuclear protected target set for UxS threat characterization. Any SSC whose loss of function would, through a credible failure chain, result in the impairment of one or more of these three safety functions is within the UxS protected target set for purposes of this annex.

The principal SSCs within the UxS protected target set are organized by safety function:

A2.1 Reactivity Control

Reactivity control SSCs include the reactor protection system (RPS) and its associated sensor and actuation infrastructure; the control rod drive mechanisms and their power supplies; and, where applicable, the emergency boron injection systems. Under normal operating conditions and during the initial shutdown phase following a military action-initiating event, reactivity control is the first safety function to be assured. The primary UxS threat to reactivity control SSCs is the disruption of the electrical and instrumentation infrastructure on which the RPS depends for sensor inputs and actuator outputs.

Above-grade instrumentation cabling, junction boxes, and sensor connections are the elements of the RPS most vulnerable to Class I FPV UAS strikes. A Class I precision strike on an RPS sensor or actuation cable run does not necessarily disable the system outright. Still, it may

introduce spurious signals or undermine the defense-in-depth provided by the redundant sensor channels on which RPS reliability depends.

A2.2 Core and Spent Fuel Pool Cooling

Cooling SSCs are the primary target category for UxS threat characterization in the nuclear context. These include: off-site power connections and yard switchgear supplying normal station power; emergency diesel generators (EDGs) and their associated fuel oil storage and supply systems; the emergency core cooling systems (ECCS) and their power supplies, motor control centers, and instrumentation; the spent fuel pool (SFP) cooling systems, including pumps, heat exchangers, piping, and instrumentation; the ultimate heat sink and its associated pump and piping infrastructure; and the emergency feedwater systems for pressurized water reactor designs.

The cooling SSC category is the highest-priority UxS protection target because the loss of cooling initiates the consequence chain that leads to core damage or SFP fuel damage. Loss of off-site power combined with loss of EDG function is the bounding initiating event for cooling degradation; this event is the direct target of a coordinated UAS campaign against the facility's electrical infrastructure. The consequence chain from loss of all cooling to core uncover or SFP thermal progression is the subject of the DBA-MA-UC-1 Use Case document.

A2.3 Radioactive Material Confinement

Confinement SSCs include the reactor containment structure and its penetrations, isolation valves, and ventilation systems; the spent fuel pool enclosure structure; the reactor coolant pressure boundary; and the spent fuel pool boundary. These structures represent the final physical barrier between radioactive material and the environment. Direct structural attack on containment by Class III or Class IV UAS warheads is within the threat envelope of state-level adversaries. The DBA-MA methodology requires that confinement integrity be maintained under the facility's design-basis UxS threat class, not only under the traditional DBA's equipment-failure postulates.

The spent fuel pool presents a specific structural vulnerability. In many reactor designs, particularly boiling water reactors (BWRs) with elevated spent fuel pool locations, the pool structure itself is accessible from above. This structural exposure makes the SFP enclosure a Category A UxS target in BWR facilities.

A3. Autonomous Safe Shutdown Condition

The Autonomous Safe Shutdown Condition (ASSC) is the nuclear sector's protected operational state for DBA-MA purposes. The ASSC is the condition in which the facility has achieved and can sustain the three nuclear safety functions — reactivity control, core and SFP cooling, and confinement of radioactive material — without dependence on off-site power, external

communications, or site access by repair and maintenance personnel. The ASSC is not a mode of normal operation; it is an engineered end state to which the facility can transition under military action conditions and sustain for a defined design-basis duration using only on-site resources.

The ASSC design basis duration — the period for which the facility must be capable of sustaining the ASSC without resupply, personnel access, or restoration of off-site systems — is established in the applicable national implementation document and is not less than the minimum period specified in the DBA-MA L2 Accident Description. This duration defines the sizing basis for on-site fuel inventories, cooling water reserves, and consumable stocks, all of which are within the UxS protected target set because their degradation shortens the ASSC duration below its design basis.

A3.1 ASSC Phases

The ASSC is structured in three phases, as defined in the DBA-MA L2 Accident Description. Understanding the phase structure is essential for applying UxS hardening requirements, because the safety functions that must be maintained and the SSCs that support them differ by phase.

Phase 1 (Immediate Response): The period immediately following the military action-initiating event, during which the facility transitions from normal operation to the ASSC. The primary safety function is reactivity control and initial core cooling. Duration: hours. UxS hardening priority: EDG survivability and RPS integrity.

Phase 2 (Sustained Decay Heat Removal): The period of sustained decay heat removal using on-site resources, without dependence on off-site power or external support. The primary safety function is the continued operation of the core and SFP cooling. Duration: days to weeks. UxS hardening priority: EDG fuel supply, ECCS and SFP cooling system integrity, and SFP water inventory.

Phase 3 (Extended Monitoring): The extended period during which the facility maintains the ASSC while awaiting the restoration of external support or the termination of the military threat condition. The primary safety function is confinement integrity and continued cooling with restored or supplemented resources. Duration: weeks to months. UxS hardening priority: replacement equipment installation security, consumable resupply route protection, and detection and monitoring system continuity.

A3.2 Local Sovereignty of Safety Functions

The local sovereignty principle requires that all SSCs credited in the ASSC safety analysis perform their safety function without dependence on external power, communications, or operator action. This principle, established in the DBA-MA L1 document, has direct implications for UxS hardening design: any SSC whose safety function depends on a system or signal that originates outside the protected zone of the facility is vulnerable to denial by UxS attack on the external dependency. The hardening design shall identify all such external dependencies and

shall either eliminate them through design modification or provide equivalent on-site backup systems that satisfy the local sovereignty requirement.

A4. Nuclear Facility Vulnerability Profiles

This section develops the generic vulnerability profiles established in DBA-UxS Sections 4.2, 5.2, and 6.2 for application to nuclear facilities. The generic profiles identify the common above-grade, waterfront, and below-waterline vulnerability patterns; this section maps those patterns to specific nuclear SSCs and identifies the consequence chain initiating events most relevant to the nuclear safety function framework.

A4.1 UAS Vulnerability Profile — Nuclear Facilities

Reactor containment structures present substantial physical resistance to Class I and Class II warheads; their direct structural compromise by UAS is not the principal concern for most facility designs. The principal concern is the above-grade support infrastructure on which containment integrity, decay heat removal, and safe shutdown depend.

Electrical yard infrastructure: Off-site power connections, yard switchgear, and main and station power transformers. Loss of off-site power is the initiating event for EDG reliance. Class II loitering munition strikes on transformers and switchgear have been demonstrated at scale in the Ukraine conflict against grid infrastructure; the same targeting logic applies to nuclear facility electrical yards. The damage mechanism is blast and fragmentation against above-grade, minimally hardened electrical equipment.

Emergency diesel generators: EDG buildings, their external fuel oil tanks, and the fuel supply piping. Class I FPV strikes can target fuel oil supply connections, breather vents, and intake and exhaust penetrations without directly penetrating the EDG building structure. Repeated Class I strikes on fuel oil supply infrastructure represent a cumulative degradation threat to the ASSC Phase 2 duration.

Spent fuel pool cooling systems: External heat exchangers, cooling pumps, and above-grade piping associated with the SFP cooling system. In many BWR designs, SFP cooling system components are located in or adjacent to the reactor building at elevations that are accessible to Class I and Class II UAS. The SFP thermal progression consequence chain, documented in DBA-MA-UC-1, is initiated by loss of SFP cooling; UAS targeting of SFP cooling components initiates this chain.

Instrumentation and control infrastructure: Above-grade cable trays, junction boxes, and sensor connections associated with the RPS and ECCS. Class I FPV platforms are specifically suited to the precision targeting of instrumentation runs that are not protected by structural hardening. The cumulative loss of redundant sensor channels through repeated Class I strikes on

instrumentation infrastructure can progressively degrade the defense-in-depth of the safety system logic.

Emergency feedwater and heat-removal systems: Above-grade components of auxiliary feedwater (PWR) or high-pressure coolant injection systems, where exposed. These systems represent the backup cooling path if primary cooling is lost; their above-grade exposure makes them secondary UAS targets after the primary electrical infrastructure.

A4.2 USV Vulnerability Profile — Nuclear Facilities

The USV threat to nuclear facilities is principally relevant to facilities sited on navigable rivers, lakes, reservoirs, or coastal water bodies that provide both the cooling water source and an unobstructed waterborne approach corridor.

Cooling water intake system: The primary nuclear facility vulnerability to USV attack. Reactor designs that depend on once-through cooling from a river, lake, or coastal water source route that water through an intake structure at or below the waterline, with an above-grade pump house and screening facilities. Loss of intake flow initiates a cooling degradation sequence that progresses through the consequence chain addressed in the DBA-MA parent documents. A Class II USV detonation at or near the intake structure produces blast overpressure, hydrodynamic shock, and fragmentation loading on the waterline-level and submerged infrastructure that conventional above-grade hardening does not address.

Waterfront logistics infrastructure: Facilities that receive fuel, consumables, or emergency equipment by waterborne transport are dependent on waterfront loading and berthing infrastructure. USV damage to this infrastructure during Phase 3 of the ASSC period can interdict the resupply of EDG fuel or barrier-replacement materials necessary to sustain the ASSC for its design-basis duration. This is the secondary USV vulnerability.

Cooling water discharge structure: The discharge path for heated cooling water returned to the water body. In facilities where discharge is routed through a submerged or waterline-level structure, USV damage to the discharge path can impair heat-rejection capability independent of the intake system's status, accelerating the thermal progression chain.

A4.3 UUV Vulnerability Profile — Nuclear Facilities

The primary nuclear facility vulnerability to UUV attack is the cooling water intake structure below the waterline. This includes the intake pipe or channel itself, the intake screening structure, and the pump suction connections. A structural breach of the intake below the waterline initiates a flooding and loss-of-cooling consequence chain that is more complex to manage than above-grade intake damage because the damage location is inaccessible for inspection and repair during an active threat period.

The secondary vulnerability is safety-significant cable or pipe runs that cross a waterway in a submerged configuration. River-crossing electrical cable runs connecting off-site power to the

facility, and submerged discharge piping, may present accessible targets for UUV explosive or sabotage operations.

BWR designs with elevated spent fuel pools have an additional UUV-relevant vulnerability: the reactor building foundation and the lower sections of the fuel-handling area structure may be accessible to below-waterline attack at facilities on rivers or lakes with sufficient water depth adjacent to the structure. Site-specific UxS vulnerability assessments for BWR facilities shall explicitly address this configuration.

A5. Nuclear-Specific Consequence Chains

The consequence threshold for nuclear facilities under DBA-MA conditions is radiological release—the uncontrolled release of radioactive material to the environment at levels that pose a public health risk. This threshold is the defining criterion against which all hardening design requirements are calibrated. The consequence chains from a UxS strike to the radiological release threshold are summarized here; a detailed analysis of the spent fuel pool thermal progression consequence chain is provided in DBA-MA-UC-1.

A5.1 Loss of Cooling Consequence Chain

The bounding consequence chain for UxS attack against nuclear facilities is: UAS strike on electrical yard infrastructure → loss of off-site power → EDG start and load assumption → if EDGs survive, ASSC Phase 1 achieved; if EDGs are subsequently attacked or fuel-depleted, ASSC Phase 2 terminates prematurely → loss of decay heat removal → core uncover (for reactor) or SFP water level decrease and boiling onset (for spent fuel pool) → fuel damage and fission product release → containment challenge and potential radiological release.

The cumulative degradation model is the more realistic threat model for this consequence chain: a single decisive strike on the EDG room is less likely than repeated Class I strikes on the EDG fuel supply, exhaust ventilation, and instrumentation over the course of a sustained UAS campaign. The hardening design shall address the cumulative scenario, including the progressive reduction of on-site fuel inventory and the sequential loss of redundant ASSC systems.

A5.2–Spent Fuel Pool Thermal Progression

The SFP thermal progression consequence chain is specifically addressed in DBA-MA-UC-1. The initiating event for UxS purposes is UAS targeting of the SFP cooling system components → loss of active SFP cooling → pool heatup and evaporation → if makeup water is not supplied, fuel uncover → zircaloy oxidation and hydrogen generation → potential fuel damage and fission product release from the SFP. This consequence chain is slower than the core cooling loss chain — measured in days rather than hours — but is more difficult to arrest once underway because the fuel damage is in an uncovered pool rather than inside a pressure vessel.

The SFP consequence chain is the priority scenario for BWR facilities because of the elevated SFP location and the large inventory of stored spent fuel that characterizes BWR fuel management cycles.

A5.3—Confinement Breach

A direct UAS attack on containment structural elements using Class III warheads constitutes a confinement-breach scenario. This scenario is less probable than the cooling-loss scenarios because containment structures are designed to meet substantial structural standards that provide inherent resistance to warhead effects at Class I and Class II levels. However, the design basis for confinement integrity under DBA-MA conditions must be evaluated against the Class III threat class for state-level adversaries. Facilities assessed as facing a state-level adversary with Class III UAS capability shall include a confinement breach structural analysis in their site-specific UxS vulnerability assessment.

A6. Nuclear-Specific Hardening Requirements

The hardening design principles established in DBA-UxS Section 8 are applied here to the nuclear SSC categories identified in Section A2. These requirements supplement the DBA-UxS Section 8 principles with nuclear-specific design-basis parameters, ASSC duration requirements, and safety-function references. The DBA-UxS Section 8 principles govern where this section does not provide more specific guidance.

A6.1 EDG Survivability

Emergency diesel generators are the highest-priority hardening target in the nuclear UxS context because their survival determines whether ASSC Phase 1 is achievable, and their sustained operability determines the duration of ASSC Phase 2. The hardening design for EDGs shall address:

Structural hardening of EDG enclosures: Against Class II blast and fragmentation loading at the design-basis standoff distance, including cumulative loading from repeated strikes at the same location.

Below-grade or hardened fuel oil storage: Dispersed storage in below-grade or hardened tanks, with the dispersal design ensuring that a single Class II strike does not eliminate more than a defined fraction of the total on-site EDG fuel inventory. The inventory sizing basis is the ASSC Phase 2 design basis duration.

Protected fuel oil supply connections: Above-grade fuel oil supply connections between storage tanks and EDGs are Class I strike targets. These connections shall be hardened, armored, or provided with hardened redundant paths sufficient to maintain fuel flow under Class I repeated strike loading.

Exhaust and intake protection: EDG exhaust and air intake penetrations are physical vulnerabilities that allow Class I warheads to cause internal damage without penetrating the structure. These penetrations shall be protected by hardened deflectors or relocated to positions not accessible to the primary approach corridors of the design-basis UAS threat.

A6.2—Spent Fuel Pool Cooling System Protection

SFP cooling system components that are above-grade or otherwise accessible to Class I and Class II UAS strikes shall be hardened consistent with DBA-UxS Section 8.1, with the design basis set at the threat class applicable to the facility. The following SFP-specific requirements apply:

Redundant SFP cooling trains: Shall be physically separated consistent with DBA-UxS Section 8.3, with separation distance sufficient that a single Class II detonation does not simultaneously affect both redundant trains.

SFP makeup water capability: A passive or gravity-driven makeup water system capable of maintaining SFP water level above the minimum safe level for the ASSC Phase 2 design basis duration, without dependence on pumps, power, or operator action. This system constitutes ASSC makeup water inventory for SFP purposes and shall be hardened against UAS strike consistent with its safety function classification.

Passive heat removal from the SFP: Where the design basis includes a passive SFP heat removal pathway — such as evaporative cooling supplemented by makeup water addition — the pathway shall be evaluated for UxS vulnerability, including the accessibility of the pool surface to Class I UAS platforms in BWR designs where the SFP is elevated and potentially accessible from above.

A6.3—Electrical Yard and Off-Site Power Interface

While off-site power itself is not an on-site SSC, the interface between the transmission system and the facility's electrical distribution — the yard switchgear, main transformers, and station service transformers — is within the facility's perimeter and within the UxS protected target set. These components cannot typically be located below grade due to their functional requirements. Still, they shall be addressed through the installation of passive kinetic barriers consistent with DBA-UxS Section 8.2, with the design-basis threat class scaled to the adversary profile.

The hardening analysis shall specifically address the recovery time for off-site power restoration following UAS damage to yard switchgear and transformers. It shall verify that the ASSC Phase 2 EDG fuel inventory is adequate to sustain cooling for the expected off-site power restoration timeline under the design-basis military action scenario.

A6.4—Cooling Water Intake System Protection

Waterfront passive kinetic barrier installations protecting the cooling water intake approach corridor shall be designed consistent with DBA-UxS Section 8.2, with the following nuclear-specific parameters:

Design-basis standoff: The barrier shall be positioned at a standoff distance from the intake structure sufficient that a Class II USV detonation at the barrier produces peak pressure and impulse loading on the intake structure below the structural damage threshold for loss of cooling water flow. This standoff shall be derived from site-specific structural analysis and is not a generic parameter.

Redundant barrier approach: The intake approach corridor shall be protected by barrier installations that provide defense-in-depth against a single barrier section failure, consistent with the nuclear safety philosophy of defense-in-depth for safety-significant barriers.

Post-strike intake integrity assessment: Following any USV strike event that engages the waterfront barrier, a subsurface inspection of the intake structure shall be conducted before concluding that the intake system has maintained its safety function. The inspection protocol shall be specified in the facility's Phase 3 ASSC operating procedures.

A7. Cross-Domain Compounding in the Nuclear Context

The sector-specific compounding scenarios called for in DBA-UxS Section 7.4 are developed here for nuclear facilities. These scenarios supplement the generic compounding frameworks in DBA-UxS Sections 7.1 through 7.3 with nuclear-specific consequence chains and SSC interdependencies.

A7.1 UAS-USV Compounding: Electrical Infrastructure and Intake Simultaneous Attack

The UAS-USV compound scenario most relevant to nuclear facilities combines UAS strikes on the electrical yard with a simultaneous USV approach to the cooling water intake. The UAS component initiates the ASSC transition by removing off-site power and forcing EDG reliance; the concurrent USV component targets the cooling-water intake system on which the ASSC cooling chain depends. If the EDG start is achieved but the cooling water intake flow is lost or severely reduced due to a USV strike, the ASSC cooling chain is broken at the heat-rejection stage, even though the EDGs are operating.

The engineering response requires that the electrical yard hardening and the waterfront barrier system be evaluated together for the compound scenario: specifically, that the waterfront barrier system satisfies the passive kinetic and local sovereignty requirements without any dependence on the electrical infrastructure that the UAS component is simultaneously targeting.

A7.2 UAS-UUV Compounding: Surveillance Degradation Before Intake Attack

In the sequential UAS-UUV scenario, UAS strikes on the facility's above-grade surveillance infrastructure — including radar antennas, camera mast structures, and the surface components of the waterborne monitoring system — degrade the facility's ability to detect the subsequent UUV approach to the cooling water intake. This scenario is particularly relevant during the Phase 3 ASSC period, when above-grade infrastructure is likely to have been damaged in the initial attack waves, and the facility's surveillance posture may be operating with reduced coverage.

The hardening design response is the requirement in DBA-UxS Section 7.2 for an independent power supply to subsurface detection systems: passive acoustic hydrophone arrays and subsurface camera systems serving the cooling water intake approach corridor shall be powered from circuits protected against above-grade UAS strikes that could degrade the surface surveillance layer.

A7.3—Cyber-Kinetic Coordination in the Nuclear Context

In the nuclear context, the cyber-kinetic scenario in DBA-UxS Section 7.3 is further developed in the companion document, *Three Threads of Cyber Security* (v1.1). The UxS-specific cyber-kinetic interaction involves using a cyber intrusion to degrade the facility's UxS detection systems immediately before the physical UAS or USV attack. The Three Threads framework establishes the cybersecurity requirements that apply to UxS detection system infrastructure in the nuclear context; this annex does not repeat that analysis but notes the interdependency for integrated design purposes.

A8. Relationship to DBA-MA Parent Documents

This annex is a child document of DBA-UxS v1.0, which is, in turn, a peer document to the DBA-MA series of instruments. The following cross-references govern the relationship of this annex to the DBA-MA series:

A8.1 DBA-UxS (v1.0)

This annex inherits the full content of DBA-UxS v1.0. All requirements established in DBA-UxS apply to nuclear facilities without modification except where this annex provides more specific or more stringent nuclear-sector requirements. Where this annex and DBA-UxS address the same requirement, the more stringent requirement governs.

A8.2 DBA-MA International Framework (L1, v1.4)

The L1 document establishes the overall DBA-MA methodology, threat categories, consequence taxonomy, and mitigation framework. This annex is within the scope of the L1 framework and does not modify any L1 requirement. The local sovereignty of safety functions principle established in the L1 document is the governing design principle for ASSC-credited SSCs, as reflected in Section A3.2 of this annex.

A8.3–DBA-MA L2 Accident Description (v2.1)

The L2 document establishes the formal accident description and design requirements for the DBA-MA. Section 4.2.5 of the L2 references passive kinetic standoff barrier systems and specifies that their design basis, threat-class scaling, and installation requirements are addressed in the UAS addendum. That reference shall be understood as referring to DBA-UxS Section 8.2 and Section A6 of this annex, effective upon adoption. Section 4.2.6 of the L2 establishes consumable inventory requirements, including on-site replacement materials for passive kinetic barriers; Section A6.1 of this annex extends those requirements to cover USV boom barrier and UUV net barrier installations.

A8.4–DBA-MA Trigger Framework (L3, v2.1)

The L3 document incorporates UAS-related trigger observables derived from the predecessor UAS addendum. The domain-specific trigger-observable framework established in DBA-UxS Section 9 — encompassing UAS, USV, and UUV indicators — shall be incorporated into the L3 document during the next L3 revision pass. Until that revision is complete, the USV and UUV trigger observables established in DBA-UxS shall be applied as supplementary guidance alongside the L3 framework.

A8.5–DBA-MA Monitoring Guidance (L4, v2.1)

The L4 document establishes post-strike inspection and restoration requirements for passive kinetic defense installations at Sections 5.6.1 and 5.6.2. Those requirements are adopted without modification and extended to cover USV boom barrier and UUV net barrier installations in addition to the UAS mesh and cable barrier systems addressed in the L4.

A8.6–Transition Architecture (DBA-MA-Transition_Architecture, v1.1)

The DBA-MA Transition Architecture (v1.1) has been updated to replace all references to the retired DBA-MA-UAS addendum with references to DBA-UxS (v1.0) and this annex. The Transition Architecture’s requirement that a facility at any transition tier must meet the UxS review cycle requirements applicable to its facility type and threat environment encompasses the full UxS review cycle requirements of DBA-UxS Section 9.

A8.7–DBA-MA Use Case Documents

DBA-MA-UC-1 (Spent Fuel Pool Thermal Progression) develops the consequence chain analysis for loss of SFP cooling under military action conditions. The UxS hardening requirements for SFP cooling systems established in Section A6.2 of this annex are calibrated to prevent the initiating events analyzed in DBA-MA-UC-1. These two documents should be read together for the SFP-specific hardening design.

A9. References

A9.1 Cross-Sector Parent Document

DBA-UxS (v1.0) — Uncrewed Systems: Threat Considerations for Critical Infrastructure: Cross-Sector Foundational Document.

A9.2–DBA-MA Series Documents

DBA-MA IAEA Framework (L1, v1.4) — Design Basis Expansion for Nuclear Power Plants Considering Acts of War: International Framework.

DBA-MA L2 (v2.1) — Design Basis Accident: Military Action Against Nuclear Facilities: Formal Accident Description and Design Requirements.

DBA-MA L3 (v2.1) — Trigger Framework for Military Action Design Basis Accident: Observable Indicators and Decision Criteria for Transition to Autonomous Safe Shutdown Condition.

DBA-MA L4 (v2.1) — Monitoring Sources and Data-Sharing Guidance for DBA-MA Trigger Observables.

DBA-MA-Transition_Architecture (v1.1) — Transition Architecture: Interim Implementation Tiers for Facilities Pending Full DBA-MA Compliance.

DBA-MA-UC-1 — Spent Fuel Pool Thermal Progression Under Military Action Conditions.

Three Threads of Cyber Security (v1.1) — Cyber Security Framework for Nuclear Facilities Under Military Action Conditions.

A9.3–IAEA References

INTERNATIONAL ATOMIC ENERGY AGENCY, Fundamental Safety Principles, IAEA Safety Standards Series No. SF-1, IAEA, Vienna (2006).

INTERNATIONAL ATOMIC ENERGY AGENCY, Safety of Nuclear Power Plants: Design, IAEA Safety Standards Series No. SSR-2/1 (Rev. 1), IAEA, Vienna (2016).

INTERNATIONAL ATOMIC ENERGY AGENCY, Nuclear Security Series No. 13, Nuclear Security Recommendations on Physical Protection of Nuclear Material and Nuclear Facilities (INFCIRC/225/Revision 5), IAEA, Vienna (2011).

A9.4—National Regulatory References

National implementation documents (US NRC, STUK, and others as applicable) are referenced in the applicable national L1 child documents. National regulatory references applicable to facility-specific hardening design analyses are identified in the site-specific UxS vulnerability assessments required under DBA-UxS Section 9.